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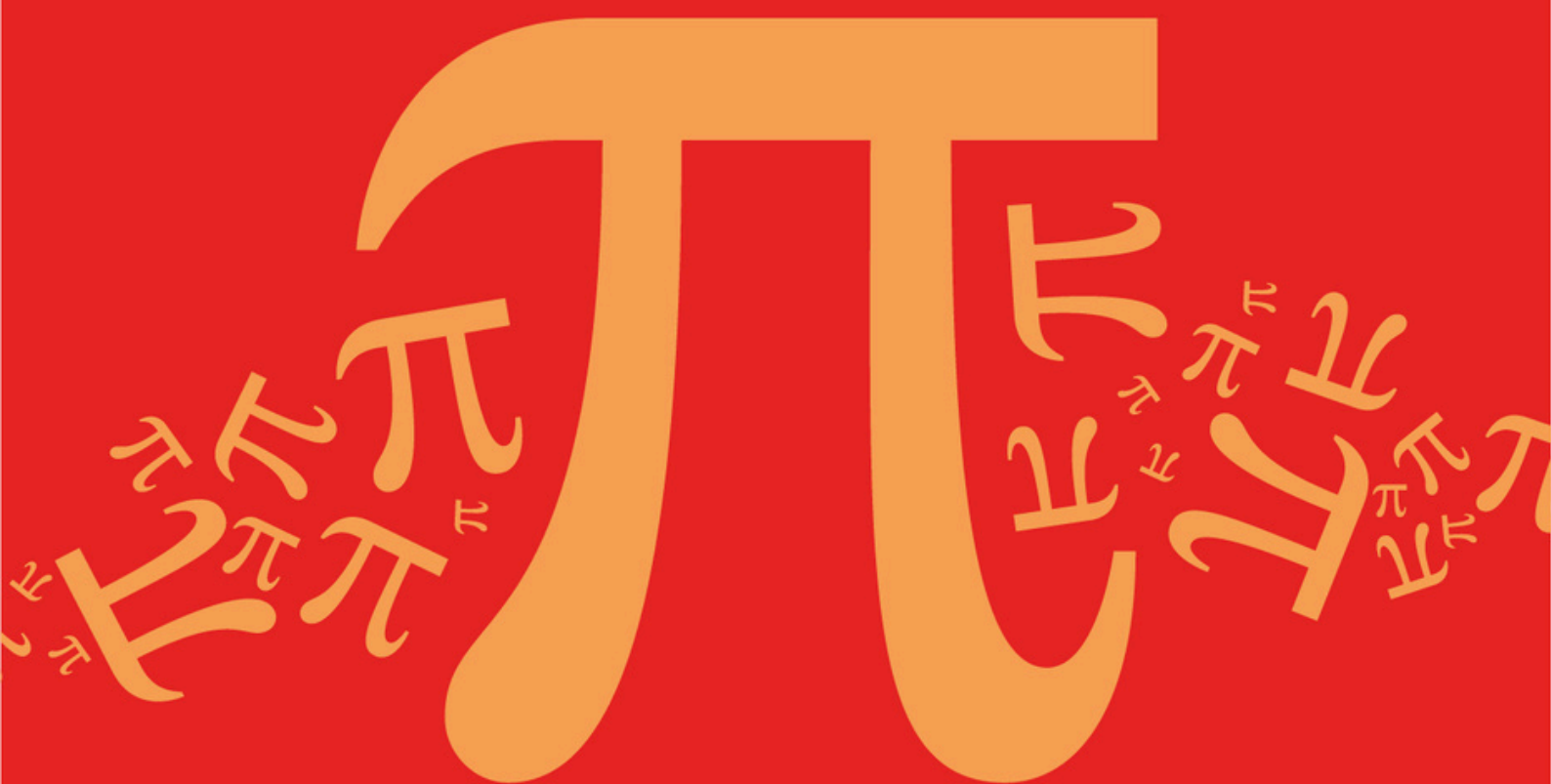
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AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C6: Singular Matrices and Inverse of a 3x3 Matrix

Mark Scheme

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses an appropriate method for finding the values of k (for example expanding appropriate determinant)	AO1.1a	M1	$\begin{vmatrix} 1 & -1 & k \\ k & -3 & 5 \\ 1 & -2 & 3 \end{vmatrix} = 0$
	Obtains a quadratic equation in k	AO1.1a	M1	$\begin{vmatrix} -3 & 5 \\ -2 & 3 \end{vmatrix} + \begin{vmatrix} k & 5 \\ 1 & 3 \end{vmatrix} + k \begin{vmatrix} k & -3 \\ 1 & -2 \end{vmatrix} = 0$
	Obtains two correct values for k	AO1.1b	A1	$1 + 3k - 5 + k(-2k + 3) = 0$ $-2k^2 + 6k - 4 = 0$ $k^2 - 3k + 2 = 0$ $(k - 2)(k - 1) = 0$ $k = 2 \text{ or } 1$
(b)	Selects an appropriate method to determine the appropriate geometrical configuration and substitutes 'their' first value of k	AO3.1a	M1	when $k = 1$ $x - y + z = 3$ $x - 3y + 5z = -1$ $x - 2y + 3z = -4$
	Eliminates one variable or uses row reduction	AO1.1a	M1	$-2y + 4z = -4$ $y - 2z = 7$
	Obtains a contradiction and makes correct deduction about the geometric configuration (must have correct value for k)	AO2.2a	R1	$y - 2z = 2; \quad y - 2z = 7$ Hence equations are inconsistent and the three planes form a prism
	Substitutes 'their' 2nd value of k into selected method to determine the appropriate geometrical configuration	AO1.1a	M1	when $k = 2$ $x - y + 2z = 3$ $2x - 3y + 5z = -1$ $x - 2y + 3z = -4$
	Obtains a consistent set of equations and makes correct deduction about geometric configuration	AO2.2a	R1	$R_2 - 2R_1 : -y + z = -7$ $R_3 - R_1 : -y + z = -7$

(must have correct value for k)			Hence equations are consistent and the three planes form a sheaf – they meet in line
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(c)	Deduces that the planes must meet in a line and hence that $k = 2$	AO2.2a	R1	$x - y + 2z = 3$ $2x - 3y + 5z = -1$ $x - 2y + 3z = -4$ $\Rightarrow -y + z = -7$ Let $z = \lambda$ Then $y = \lambda + 7$ and $x = 3 + y - 2z$ $= 3 + \lambda + 7 - 2\lambda$ $= -\lambda + 10$
	Selects method to find solution: For example, sets one variable $= \lambda$, substitutes and attempts to find other variables in terms of λ	AO1.1a	M1	ALT $\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$
	Fully states correct solution CAO	AO1.1b	A1	$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 10 \\ 7 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$
Total 11 marks				

Q2.

(a)
$$\begin{vmatrix} -2 & 1 & 2k \\ -1 & 1 & k+1 \\ 2 & k-1 & 1 \end{vmatrix} = -2 \begin{vmatrix} 1 & k+1 \\ k-1 & 1 \end{vmatrix} + 1 \begin{vmatrix} 1 & 2k \\ k-1 & 1 \end{vmatrix} + 2 \begin{vmatrix} 1 & 2k \\ 1 & k+1 \end{vmatrix}$$

correctly expanding by any row or column

M1

$$= -2[1 - (k+1)(k-1)] + [1 - 2k(k-1)] + 2[k+1 - 2k]$$

correct unsimplified expansion of 2×2 determinants

A1

$$= -2[1 - k^2 + 1] + [1 - 2k^2 + 2k] + 2[1 - k]$$

$$= -4 + \cancel{2k^2} + 1 - \cancel{2k^2} + \cancel{2k} + 2 - \cancel{2k}$$

$$= -1$$

-1 obtained

A1cso

either all k's cancel or independent of k etc

comment required (must score previous 3 marks)

E1

4

Alternative:

$$\begin{vmatrix} -2 & 1 & 2k \\ -1 & 1 & k+1 \\ 2 & k-1 & 1 \end{vmatrix} \quad \begin{array}{l} r_1 \rightarrow r_1 - 2r_2 \\ r_3 \rightarrow r_3 + 2r_2 \end{array}$$

$$= \begin{vmatrix} 0 & -1 & -2 \\ -1 & 1 & k+1 \\ 0 & k+1 & 2k+3 \end{vmatrix} \quad r_3 \rightarrow r_3 + (k+1)r_1$$

correctly expanding by any row or column after row operations

(M1)

$$= \begin{vmatrix} 0 & -1 & -2 \\ -1 & 1 & k+1 \\ 0 & 0 & 1 \end{vmatrix} = -1$$

correct expansion unsimplified

(A1)

-1 obtained

(A1)

either all k's cancel or independent of k etc

comment required

(E1)

(4)

(b) (i) Identifying that $k = 3$

$$k = 3$$

B1

Value of determinant $\neq 0$ (or $= -1$ etc) therefore equations are consistent

ft answer (a) if $0 \Rightarrow$ inconsistent

E1F

2

(ii) 3 planes intersect in a unique point

B1

1

[7]

Q3.

(a)
$$\left[\begin{array}{ccc|c} 2 & -1 & -1 & 3 \\ 1 & 2 & -3 & 4 \\ 2 & 1 & a & b \end{array} \right]$$

$$\begin{array}{l} r_3 \rightarrow r_3 - r_1 \\ r_2 \rightarrow 2r_2 - r_1 \end{array} \quad \left[\begin{array}{ccc|c} 2 & -1 & -1 & 3 \\ 0 & 5 & -5 & 5 \\ 0 & 2 & a+1 & b-3 \end{array} \right]$$

Correct row operations used to create two zeros in first column – coefficients must be correct

M1

$$r_3 \rightarrow 5r_3 - 2r_2 \quad \left[\begin{array}{ccc|c} 2 & -1 & -1 & 3 \\ 0 & 5 & -5 & 5 \\ 0 & 0 & 5a+15 & 5b-25 \end{array} \right]$$

Use of row operations to create third zero in second column or compare ratios of coefficients in rows 2 and 3

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M1

No unique solution: $5a + 15 = 0$

$$a = -3$$

Solves equation with coefficient of $z = 0$ or equation formed from comparison of ratios (eg $a + 1 = -2$) $a = -3$ is a printed answer

A1

3

Alternative 1:

$$2x - y - z = 3 \quad (1)$$

$$x + 2y - 3z = 4 \quad (2)$$

$$2x + y + az = b \quad (3)$$

$$(1) + (3) \Rightarrow 4x + (a-1)z = b+3 \quad (4)$$

$$2(3) - (2) \Rightarrow 3x + (2a+3)z = 2b-4 \quad (5)$$

Correct elimination of 1 variable.

Coefficients must be correct.

(M1)

$$4(5) - 3(4) \Rightarrow (5a + 15)z = 5b - 25$$

Correctly reduce to one equation with a, b

(M1)

$$5a + 15 = 0$$

$$a = -3$$

Solves equation with coefficient of $z = 0$

(A1)

(3)

Alternative 2:

$$\text{Solve } \begin{vmatrix} 2 & -1 & -1 \\ 1 & 2 & -3 \\ 2 & 1 & a \end{vmatrix} = 0$$

$$2 \begin{vmatrix} 2 & -3 \\ 1 & a \end{vmatrix} - 1 \begin{vmatrix} -1 & -1 \\ 1 & a \end{vmatrix} + 2 \begin{vmatrix} -1 & -1 \\ 2 & -3 \end{vmatrix} = 0$$

Correct expansion by row or column

(M1)

$$2(2a + 3) - (-a + 1) + 2(5) = 0$$

Correct expansion of 2 by 2 determinants

(M1)

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$$5a + 15 = 0$$

$$a = -3$$

Solves equation with determinant = 0

(A1)

(3)

(b) Either $5b - 25 \neq 0$ or $5b - 25 = 0$

Sets their constant $\neq 0$ (or 0)

M1

Inconsistent $b \neq 5$

CSO (accept $b > 5, b$

A1

2

[5]

Q4.

(a) $\text{Det}(AB) = (4)(1) - (1)(1) = 3 \neq 0$

Must state non-zero or $\neq 0$

B1

1

(b) Matrix of cofactors $\begin{bmatrix} 0 & 0 & 3 \\ 4 & -1 & 8-4k \\ -1 & 1 & k-8 \end{bmatrix}$

Attempt at matrix of cofactors – at least five correct entries

M1

Fully correct matrix of cofactors

A1

$$(AB)^{-1} = \frac{1}{3} \begin{bmatrix} 0 & 4 & -1 \\ 0 & -1 & 1 \\ 3 & 8-4k & k-8 \end{bmatrix}$$

Their cofactor matrix transposed correctly

M1

At least five correct = A1 (exclude effect of determinant)

All entries fully correct = A2

A2,1

5

(c) $(AB)^{-1} = B^{-1}A^{-1}$

$\Rightarrow B^{-1} = (AB)^{-1}A$

$$B^{-1} = \frac{1}{3} \begin{bmatrix} 0 & 4 & -1 \\ 0 & -1 & 1 \\ 3 & 8-4k & k-8 \end{bmatrix} \begin{bmatrix} k & 6 & 8 \\ 0 & 1 & 2 \\ -3 & 4 & 8 \end{bmatrix}$$

Use of $(AB)^{-1}$ and A multiplied

M1

Correct order of multiplication

A1

$$= \frac{1}{3} \begin{bmatrix} 3 & 0 & 0 \\ -3 & 3 & 6 \\ 24 & -6 & -24 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 2 \\ 8 & -2 & -8 \end{bmatrix}$$

All correct = A2

5+ correct = A1 (exclude effect of determinant)

NB – if an attempt is made to find B by setting up a

system of simultaneous equations then

M1 – 9 correct equations used

$$A1 \text{ for } B = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 2 & \frac{1}{2} \\ \frac{3}{2} & -\frac{1}{2} & -\frac{1}{4} \end{pmatrix}$$

Final A2 as above

A2,1

4

[10]

Q5.

(a)

$$\begin{vmatrix} 2 & n & 1 \\ 3 & -1 & n \\ -1 & 7 & 1 \end{vmatrix}$$

Expanding the det. of the coeff. mtx.

M1

Setting it = 0

M1

Obtaining & solving a quadratic eqn. in n

M1

$$0 = n^2 + 17n - 18 = (n + 18)(n - 1) \Rightarrow n = 1, -18$$

CSO

A1

4

(b) $n = 1$ gives $2x + y + z = 5$

$$3x - y + z = 1$$

$$-x + 7y + z = 1$$

fit their chosen integer n

B1

Eliminating one variable from a pair of equations, twice

M1

e.g. $\textcircled{2} - \textcircled{1} \Rightarrow x - 2y = -y = -4$

A1 ft

and $(2) - (3) \Rightarrow 4x - 8y = 0$

A1 ft

Inconsistency clearly demonstrated from fully correct working

E1

3 planes have no common intersection (or form a Δ^r prism)

Also ft "3 planes meet in a common line" or

"3 planes form a sheaf" if consistency conclusion made

B1 ft

6

[10]

Q6.

(a) (i) $\mathbf{A}^T = \begin{bmatrix} k & 2 & 1 \\ 1 & k & 2 \\ 2 & 1 & k \end{bmatrix}$

B1

Good multiplication attempt at $\mathbf{A} \mathbf{A}^T$

If they multiply $\mathbf{A}^T \mathbf{A}$ instead, they can score B1 M1 A0 A0 A1 A1

A1A1

©2025 Exam Papers Practice. All rights reserved. M1

$$= \begin{bmatrix} k^2 + 5 & 3k + 2 & 3k + 2 \\ 3k + 2 & k^2 + 5 & 3k + 2 \\ 3k + 2 & 3k + 2 & k^2 + 5 \end{bmatrix}$$

Main diagonal correct

A1

All others correct

A1

$$k = -\frac{2}{3} \quad m = 5\frac{4}{9} \quad \text{or} \quad \frac{49}{9}$$

6

(ii) $\begin{bmatrix} -\frac{2}{3} & 1 & 2 \\ 2 & -\frac{2}{3} & 1 \\ 1 & 2 & -\frac{2}{3} \end{bmatrix}^{-1} = \frac{9}{49} \begin{bmatrix} -\frac{2}{3} & 2 & 1 \\ 1 & -\frac{2}{3} & 2 \\ 2 & 1 & -\frac{2}{3} \end{bmatrix}$

1

ft their m and their k

B1

Accept $\frac{9}{49} \begin{bmatrix} k & 2 & 1 \\ 1 & k & 2 \\ 2 & 1 & k \end{bmatrix}$ since the value of k is now known,
but not just $\frac{9}{49} \mathbf{A}^T$

Decimal version (correct to at least 3sf) is also ok:

$$\begin{bmatrix} -0.122 & 0.367 & 0.184 \\ 0.184 & -0.122 & 0.367 \\ 0.367 & 0.184 & -0.122 \end{bmatrix}$$

1

- (b) (i) $\det \mathbf{A} = k^3 - 6k + 9$
Good attempt (cubic)

M1
A1

2

- (ii) $\det \mathbf{A} = (k + 3)(k^2 - 3k + 3)$
Factorisation attempt

M1

$$k = -3$$

A1

$$\Delta = 9 - 12$$

B1

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Special Cases

$k = -3$ but no working

(B1)

For all 3 roots, -3 and $\frac{3+i\sqrt{3}}{2}$ given
with no supporting working

(B3)

3

- (iii) Replacing k by $(k - 7)$
NOT just in the determinant form (ie starting again)

M1

$k = 4$ (however obtained)
ft their previous $k + 7$

A1

2

Q7.

(a) (i) $\mathbf{X}^2 = \begin{bmatrix} 23 & 1 & -3 \\ 2 & 24 & 3 \\ -4 & 2 & 19 \end{bmatrix}$
 ≥ 5 correct for the M

M1

All 9 correct for the A

A1

$\mathbf{X}^2 - \mathbf{X} = 20\mathbf{I}$ i.e. $k = 20$
 Shown legitimately

A1

3

(ii) Mult⁹. $\mathbf{X}^2 - \mathbf{X} = 20\mathbf{I}$ by $\mathbf{X}^{-1}$

M1

Re-arranging $\mathbf{X} - \mathbf{I} = 20 \mathbf{X}^{-1}$

to get $\mathbf{X}^{-1} = \frac{1}{20}(\mathbf{X} - \mathbf{I})$

Legitimately

A1

2

(b) $\mathbf{X}^{-1} = \frac{1}{20} \begin{bmatrix} 2 & 1 & -3 \\ 2 & 3 & 3 \\ -4 & 2 & -2 \end{bmatrix}$

Noted or used

B1

$(\mathbf{XY})^{-1} = \mathbf{Y}^{-1} \mathbf{X}^{-1}$

Incl. attempt at the multn.

M1

$\begin{bmatrix} 6 & 3 & -9 \\ 2 & -1 & 1 \\ 2 & 3 & 3 \end{bmatrix}$

A1

3

[8]

Q8.

(a)
$$\begin{bmatrix} 32 - 4t & 0 & 0 \\ 98 - 2t^2 & 4t - 24 & 2t - 14 \\ 28 - 4t & 0 & 4 \end{bmatrix}$$

*Decent attempt at **AB***

M1

≥ 5 correct

A1

all correct

A1

$t = 7$

*Allow this ft if only 1 or 2 elements of **AB** incorrect*

A1

$\mathbf{AB} = 4\mathbf{I}$

*Must be from **AB** completely correct*

A1

5

(b)
$$\mathbf{A}^{-1} = \frac{1}{4} \begin{bmatrix} 115 & -4 & -1 \\ -14 & 4 & 2 \\ 17 & -4 & -3 \end{bmatrix}$$

*ft 1/det if related to **B***

B1

CAO; must be “deduced”

$\frac{1}{4} \mathbf{B}$
NB “ $\frac{1}{4}$ ” scores B1 only

B1

2

[7]

Q9.

(a) $\text{Det } \mathbf{A} = k + 3 + 12 - 4 - 9 - k = 2$

M1

CAO

A1

2

(b)
$$\mathbf{A}^{-1} = \frac{1}{\text{Det } \mathbf{A}} (\text{adj } \mathbf{A})$$

Correct use of the determinant (any value)

B1

Attempt at matrix of cofactors

M1

$$= \frac{1}{2} \begin{bmatrix} k-9 & 3-k & 2 \\ 12-k & k-4 & -2 \\ -1 & 1 & 0 \end{bmatrix}$$

Use of transposition **and** signs

M1

At least 5 entries correct (even if 2nd M1 not earned)

A1

CAO – ft det only

A1

5

[7]

Q10.

(a) (i) $\det \mathbf{P} = 4a + 6 + 4 + a = 5a + 10$

M1A1

2

(ii) When $a = 3$, $\det \mathbf{P} = 25$
ft

B1F

1

(iii) Setting their $\det \mathbf{P} = 0 \Rightarrow a = -2$

M1

ft

A1F

2

(b) (i) $\mathbf{P}^{-1} = \frac{1}{25} \mathbf{Q}$

B1

1

(ii) $(\mathbf{PQ})^{-1} = (25 \mathbf{I})^{-1} = \frac{1}{25} \mathbf{I}$

M1A1

2

Or $(\mathbf{PQ})^{-1} = \mathbf{Q}^{-1} \mathbf{P}^{-1}$

Ignore $(PQ)^{-1} = P^{-1} Q^{-1}$ if they can make it work

(M1)

$$= Q^{-1} \cdot \frac{1}{25} Q = \frac{1}{25} I$$

(A1)

(2)

(iii) $\det PQ = \det (25 I) = 25^3$ or 15625

M1A1

$\det PQ = \det P \cdot \det Q$

Used

M1

$$\Rightarrow 25^3 = 25 \det Q$$

$$\Rightarrow \det Q = 25^2 \text{ or } 625$$

A1

4

[12]

Q11.

(a) Setting $\det P (2t - 6 + 2 - 3t + 8 - 1 = 3 - t) = 0$

M1

$$\Rightarrow t = 3$$

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A1

2

(b) (i)
$$\begin{bmatrix} 6 & 0 & 0 \\ -7t - 21 & 3 - t & 15 + 5t \\ 0 & 0 & 6 \end{bmatrix}$$

Multiplication attempt with ≥ 3 correct elements

M1

Any one row or column

A1

All correct

A1

3

(ii) When $t = -3$, $PQ = 6I$

The 6 may be implicit

B1 B1

2

$$(iii) \quad \mathbf{Q}^{-1} = \frac{1}{6} \mathbf{P} = \frac{1}{6} \begin{bmatrix} 2 & 1 & 1 \\ 1 & -3 & -2 \\ 3 & 2 & 1 \end{bmatrix}$$

Allow for $\frac{1}{6} \mathbf{P}$ or $\frac{1}{36} (6\mathbf{P})$

B1

1

(c) Enlargement (centre O) sf 6

M1 A1

2

[10]

Examiner reports

Q2.

Expansions of determinants are well understood, although there were often sign errors with the terms 1 , -1 , $k + 1$ and $k - 1$. Several students failed to comment on their result to show they had understood what was meant by 'independent of k '. Many students failed to see the link between the determinant in part (a) and the rest of the question, with some starting again at each part. Furthermore, it is clear that students do not understand the concept of consistency of a system of equations. Many responses were seen that stated 'consistent' and gave a prism as the configuration or 'inconsistent' and sheaf.

Q3.

The most successful method in part (a) was to find the determinant and set it to equal zero. The least successful approach was to try to combine equations in a more traditional manner. Students simply got themselves lost in the algebra and could not complete the method. The row operation method of elimination was rarely seen but often proved successful and helped with part (b). Part (b) illustrated again that students are not confident or clear about consistent and inconsistent equations readily getting them mixed up.

Q4.

A good response was received to this question. In part (a) it is not sufficient to give the determinant as 3 and deduce that the matrix is non-singular. The key point to secure the mark is to state that $3 \neq 0$ and then make the deduction. Inverse matrices are well understood, marks were dropped here for the standard types of errors with signs or miscalculation of minor determinants. Part (c) proved more differentiating. A minority of students were not aware of the result $(AB)^{-1} = B^{-1}A^{-1}$ and hence set off on a trail to find B^{-1} by other, often invalid, means.

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Q5.

This was another generally popular and successful question for candidates. Most attempts scored all 4 marks on part (a) and most of those for part (b). However, the approaches taken often provided further evidence that the candidature, in general, was not at the appropriate level of mathematical maturity required for taking their MFP4 paper at this time. Having supplied three equations with a common coefficient of 1 for z , easing the work needed to establish the inconsistency of the system, most candidates preferred to work with the x 's and y 's instead, which just took a little longer and led more often to the introduction of numerical errors. Also, it was often the case that inconsistency was "established" on the basis that two constants weren't equal, when comparing a couple of equations, but candidates' lack of care in the arithmetic frequently meant that one or other of these two constants wasn't the correct one, which invalidated their conclusion and lost them a mark.

Q6.

Apart from the few who did not seem to know what a transpose matrix was, this was a popular and high-scoring question, even for those with relatively low scores on the paper as a whole. The only really disturbing aspect of the work on display was that in part (b)(ii) so few students seemed to have any idea how to go about finding a linear factor of a fairly simple cubic and then factorising to find the accompanying quadratic factor, which is C1

work. Most students who gained the answer $k = -3$ appeared to get it from a calculator, and very few indeed could use it to find the required quadratic factor; fewer still were able to reason convincingly that this quadratic factor had no real roots, which again is C1 work.

In the final part of the question, many students failed to realise that k was now being replaced in the given matrix **A** by $k - 7$, so that all they had to do was to add 7 to their previous value of k for two straightforward marks. Almost all students who correctly found the answer for this new k did so by going back, working out a cubic characteristic equation (with some large numbers for coefficients), and using a calculator to solve it for them. This was, somewhat reluctantly, awarded the marks if the answer was correct. Most of those students who did not get the correct answer also used this method, but any incorrect working prevented their calculator helping them out and so they gained no marks.

Q7.

This was another popular question, in that responses were generally at least partially successful. Even so, around half of all candidates failed to realise that the “hence” in part (a)(ii) meant that they were supposed to work algebraically with the result of part (a)(i) rather than go through the lengthy approach of finding $\mathbf{X}^{-1}$ directly. About the same proportion incorrectly opted for $(\mathbf{XY})^{-1} = \mathbf{X}^{-1}\mathbf{Y}^{-1}$. In addition to all these errors, a lot of marks were lost due to carelessness; mistakes made in calculating $\mathbf{X}^2$ were often not corrected, even when the candidates clearly failed to arrive at a multiple of **I** in part (a)(i).

Then, despite being told that $\mathbf{X}^{-1}$ could be found by $\frac{1}{20}(\mathbf{X} - \mathbf{I})$ in part (b), many made a slip

somewhere in the working, and the correct $(\mathbf{XY})^{-1}$ appeared far less often than should have been the case, given the information given in the question.

Q8.

The matrix multiplication was generally handled very well, apart from a small but noticeable number of candidates who insisted on attempting **BA** despite the question. The only other obstacle to a completely correct part (a) was found amongst those candidates who had the odd one or two incorrect elements, which they had failed to notice and correct due to a lack of a check, meaning that $t = 7$ did not actually give $4\mathbf{I}$ consistently for their **AB**. As with question 2, the word “deduce” in part (b) was almost totally ignored, and alternative methods for finding an inverse, often taking up lots of time, received no credit.

Q9.

Almost all candidates did very well at finding the inverse matrix using the ‘transposed matrix of co-factors’ approach, and only a small proportion forgot the alternating signs and/or the transposition.

Q10.

This was the last question on the paper because, since it involves a 3×3 matrix, it was potentially the most time-consuming question. However, candidates found the structure of the question very helpful.

Q11.

This proved to be a profitable question for many candidates. In particular, the

multiplication of two 3×3 matrices was handled very competently indeed, even though algebraic terms were involved. The odd error arising here did not prevent candidates from doing good work in the later parts of the question. They were not required to check that all entries of the product matrix gave consistent values for both t and k .



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